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Quantitative evaluation and investigation of shielding coefficient and magnetic noise in stacked nanocrystalline magnetic shielding system

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A B S T R A C T

The utilization of nanocrystalline magnetic shielding can provide a low-noise weak magnetic environment for various ultra-high sensitivity measurements. Their interlayer insulation properties effectively impede eddy currents, thereby ameliorating eddy current losses and diminishing magnetic noise within the magnetic shielding structure. However, the nanocrystal's thin-layered structure and stacking process greatly impact its effective magnetic permeability and conductivity. Hence, this article presents a quantitative evaluation and research methodology for the shielding coefficient and magnetic noise in nanocrystal shielding systems, considering the influence of layer stacking. Firstly, the complex magnetic permeability of nanocrystal annulus with different stacking coefficients is measured, demonstrating a linear relationship. Secondly, based on the test results, a cubic nanocrystal shielding effectiveness model is constructed and validated against finite element results. Finally, an improved magnetic noise calculation model is analyzed and developed for square nanocrystal shielding bodies by utilizing the linear relationship and considering the truncation of eddy currents in different directions within the stacked nanocrystal shielding layer. The accuracy of the model is verified through experimental validation.

1. Introduction

Atomic gyroscopes [1], magnetometers [2], and other highly sensitive atomic sensors [3] have a wide range of applications in fundamental physics research [4], aerospace [5], and biomedical fields [6] due to their high-resolution measuring capabilities. In biomedical research, the presence of calibration errors in optically-pumped magnetometers (OPMs) [7] can lead to the obfuscation of biological magnetic fields, causing measurement inaccuracies when measuring magnetic fields of the heart [8], brain [9], muscles, and cells [10]. High shielding and low noise are essential for ensuring the proper functioning of high-precision atomic sensors; however, the intrinsic noise of the materials that make up the magnetic shielding system remains a major source of constraint on their sensitivity and accuracy. In the future, nanocrystalline shielding material holds promise for providing a low residual magnetism and low magnetic noise calibration environment.

Traditional magnetic shielding devices are composed of permalloy layers and aluminum layers, with the former being a high-permeability material that shields low-frequency external magnetic fields using the magnetic shunt effect principle and the latter being a high-conductivity material that shields high-frequency magnetic fields using the eddy current effect. However, the high conductivity and high permeability characteristics of these shielding devices lead to the presence of Johnson noise and thermal magnetic noise, which limit the sensitivity of atomic sensors [11]. Therefore, to improve the shielding effect while reducing the magnetic noise caused by eddy currents and hysteresis, researchers are constantly exploring the application scenarios and scope of new materials, such as superconducting materials, ferrites, and iron-based nanocrystals. Superconducting materials [12,13] can maintain a constant shielding coefficient below 10 Hz and reduce noise by a hundredfold compared to permalloy, but they require extremely low temperatures and have high manufacturing and usage costs. The extremely low electrical conductivity of ferrites makes them an ideal shielding material for providing a low-noise environment for many atomic sensors [14], but their texture is thick, stress-sensitive, and

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difficult to demagnetize due to their subferromagnetic ceramic properties. Nanocrystalline materials, in comparison to other soft magnetic materials, possess high saturation magnetization, low coercivity, high magnetic permeability, and high electrical resistivity [15]. Furthermore, their unique nanoscale structure and relatively thin laminations can block eddy currents, improve eddy current losses, and thus serve as a new material for manufacturing magnetic shields to reduce internal noise within the shields [16].

Currently, research on nanocrystalline magnetic shielding devices has been limited to barrel-shaped structures, overlooking the impact of laminations on magnetic noise assessment. Simultaneously, the higher spatial utilization of a cubic hexahedral enclosure presents new challenges for magnetic noise modeling and assessment. Fu et al. [17] have demonstrated the potential of nanocrystalline material in low-noise magnetic shielding for SERF magnetometers by applying it to the inner layer of the shielding. Additionally, Liu et al. [18] have experimentally proven that nanocrystalline magnetic shielding exhibits lower thermal-magnetic noise and better temperature stability compared to amorphous shielding. However, the aforementioned studies only focused on magnetic noise testing of nanocrystalline barrels and did not consider the influence of lamination structures on eddy current noise from the material itself.

In this study, we employed a quantitative evaluation method to consider the influence of layered structure on permeability. Additionally, we introduced the concept of layered electrical conductivity and conducted magnetic noise modeling for a regular hexahedron laminated nanocrystalline magnetic shielding device. Compared to conventional methods for evaluating shielding effectiveness and magnetic noise that are only applicable to conventional dense materials, this improved calculation method for the shielding coefficient and magnetic noise in nanocrystal magnetic shielding systems takes into account the interlayer insulation achieved between the laminates to block eddy currents [19,20]. The remaining parts of this paper are organized as follows: In Section 2, the complex magnetic permeability of nanocrystal annulus with different laminate coefficients is tested, and a model is summarized for the variation of the effective magnetic permeability and electrical conductivity with the laminate coefficient. In Section 3, based on the model of effective magnetic permeability, a model for the shielding effectiveness of nanocrystal cubic shields is constructed, and its effectiveness is validated by comparison with finite element results. Section 4 analyzes and improves the magnetic noise calculation model applicable to square nanocrystal magnetic shields and verifies the accuracy of the model through experiments.

2. The model of permeability and conductivity considering lamination coefficient

2.1. The model of permeability

The complex magnetic permeability is the evaluation criterion for the shielding performance of nanocrystalline shielding materials. The complex magnetic permeability is greatly influenced by the laminated process (interlayer insulation) of the nanocrystals. Therefore, measuring and studying the laminated structures of nanocrystals with different laminate coefficients is necessary. Traditional measurements of the magnetic permeability of nanocrystal annulus mainly focus on experiments conducted on bare nanocrystal rings without interlayer insulation treatment. However, these measurement results cannot be applied to evaluating the shielding effectiveness of actual laminated nanocrystal magnetic shielding devices.

To consider the effect of lamination, the traditional toroidal sample is improved by introducing non-magnetic materials between the layers, thereby changing its laminate coefficient and measuring its complex magnetic permeability. As shown in Fig. 1, by winding a coil with 60 turns around the circular nanocrystalline sample, the measured variables are resistance and inductance. Based on the “coil method”, the

Fig. 1. Schematic diagram of testing the complex magnetic susceptibility of the ring using an LCR meter.

complex magnetic permeability can be derived and calculated [21,22]:

$$\mu' = \frac{l_e}{\mu_0 A_e N_{\text{coil}}^2} L \tag{1}$$

$$\mu'' = \frac{l_e}{2\pi f \mu_0 A_e N_{coil}^2} \left(R - R_w\right) \tag{2}$$

where μ' and μ'' are the real part and imaginary part of the relative complex permeability respectively, L and R represent the measured inductance and resistance, respectively. R_w is the resistance of the wire, A_e is the cross-sectional area, N_{coil} is the number of turns, f is the frequency, and l_e is the equivalent length:

$$l_e = \frac{\pi\left(d_o - d_i\right)}{\ln\left(d_o/d_i\right)} \tag{3}$$

where d_o and d_i represent the outer and inner diameters of the toroidal sample, respectively.

In a multi-layer magnetic shielding system, the nanocrystalline magnetic shielding layer is typically located in the innermost layer. Since the innermost layer experiences a low-frequency weak magnetic environment, the complex magnetic permeability used to calculate the nanocrystalline magnetic noise and shielding coefficient should be measured under weak low-frequency excitation. However, due to the limited accuracy range of the LCR meter (16089B) at low-frequency weak magnetic excitations, directly measuring the complex magnetic permeability is challenging. Therefore, the Rayleigh relationship needs to be applied to fit and predict the complex magnetic permeability. Under a fixed frequency of 20 Hz, the LCR meter is used to apply a driving voltage to the toroidal coil. When the driving voltage is sufficiently small, the initial complex magnetic permeability can be linearly related to the amplitude magnetic permeability. Thus, in the case of a small-amplitude external magnetic field, the magnetization curve in the Rayleigh region [23] is described by:

$$\mu_a = \mu_i + \eta H \tag{4}$$

where μ_a represents the amplitude magnetic permeability, μ_i represents the initial magnetic permeability, and η represents the Rayleigh constant.

In order to validate the effect of the laminating process on the complex magnetic permeability, experiments were conducted using nanocrystalline toroidal samples with varying laminating coefficients. The laminating process involved the application of a silicon dioxide sol–gel coating as an interlayer insulation between the nanocrystalline ribbon layers. By keeping the thickness of the nanocrystalline ribbon

Table 1
Measured permeability at different frequencies.

Frequency (Hz)	Test ring 1(e = 0.49)		Test ring 2(e = 0.54)		Test ring 3(e = 0.59)		Test ring 4(e = 0.63)		Test ring 5(e = 0.65)		Test ring 6(e = 0.68)	
	μ'	μ''	μ'	μ''	μ'	μ''	μ'	μ''	μ'	μ''	μ'	μ''
20	17 810	86	19 361	95	20 740	106	22 763	114	23 512	120	24 811	128
30	17 731	88	19 354	90	20 734	102	22 769	112	23 523	119	24 823	125
40	17 815	79	19 362	91	20 747	104	22 759	115	23 519	121	24 815	130
50	17 823	81	19 344	96	20 739	101	22 771	111	23 515	118	24 822	122
60	17 803	76	19 378	98	20 756	99	22 761	118	23 528	119	24 825	129
70	17 795	89	19 366	92	20 736	104	22 765	119	23 518	122	24 816	127
80	17 813	85	19 387	95	20 748	102	22 758	114	23 525	117	24 817	126
90	17 822	83	19 355	91	20 739	105	22 765	112	23 526	122	24 827	125
100	17 808	89	19 374	93	20 745	106	22 767	113	23 514	121	24 808	128

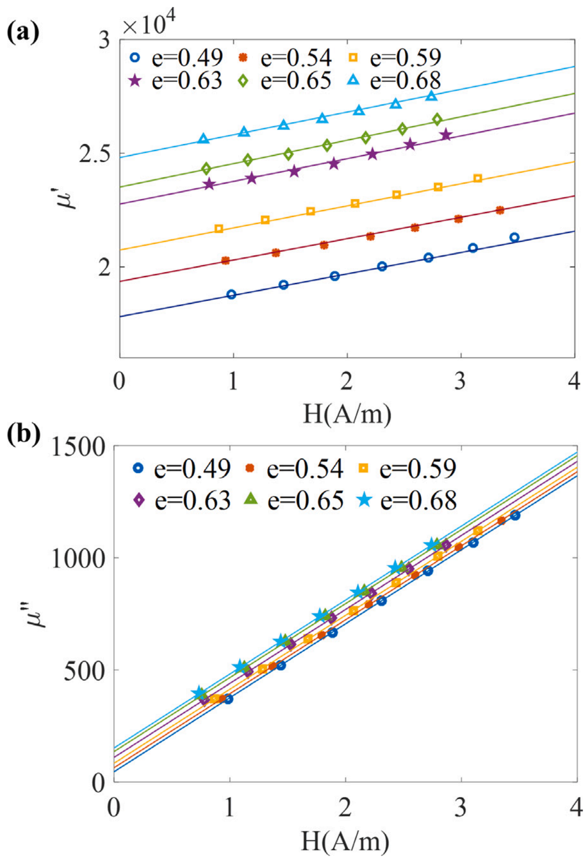

Fig. 2. Fitted curves of the complex permeability of iron-based nanocrystalline rings. (a) Real part of permeability. (b) Imaginary part of permeability.

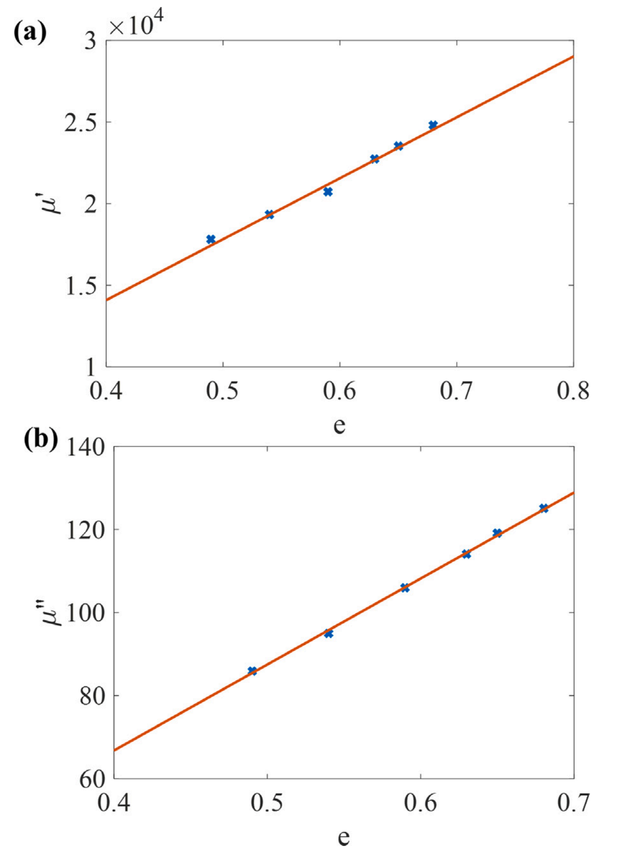

Fig. 3. Fitted curves of the complex permeability of iron-based nanocrystalline rings concerning the stacking factor. (a) Real part of permeability concerning the stacking factor. (b) Imaginary part of permeability concerning the stacking factor.

consistent, different laminating coefficients were achieved by varying the thickness of the interlayer insulation coating. The effective thickness of the magnetic shielding layer on the nanocrystalline test ring was denoted as d_1, while the actual thickness was denoted as d_2. The laminating coefficient was defined as $e(e = d_1/d_2)$. Using an LCR meter (16089B), the complex magnetic permeability was measured for the nanocrystalline samples with different laminating coefficients under the application of the same driving voltage to the coil. The measured results of the complex magnetic permeability corresponding to different driving fields are shown in Fig. 2.

As shown in Table 1, the magnetic permeability of nanocrystalline materials remains almost constant within the low-frequency range of 20–100 Hz. Fig. 2(a) and (b) represent the real and imaginary parts of the complex magnetic permeability, respectively, at different magnetic field strengths. A linear fitting was performed based on Eq. (4). The linear curve exhibited a consistent upward or downward trend as the laminating coefficient varied. Fig. 3 represents the complex magnetic permeability fitting results at zero driving field from Fig. 2. By performing a linear fitting on the data in Fig. 3 as well, a linear relationship between the complex magnetic permeability and the laminating coefficient can be obtained:

$$\mu' = 37340 \cdot e - 848.8 \tag{5}$$

$$\mu'' = 214.6 \cdot e - 20 \tag{6}$$

2.2. The model of conductivity

The magnitude of magnetic noise in nanocrystalline shielding material depends on the complex magnetic permeability and electrical conductivity. The previous text has elaborated on the relationship between the complex magnetic permeability and the laminating coefficient. The interlayer insulation process in nanocrystalline materials also affects the effective electrical conductivity by suppressing eddy

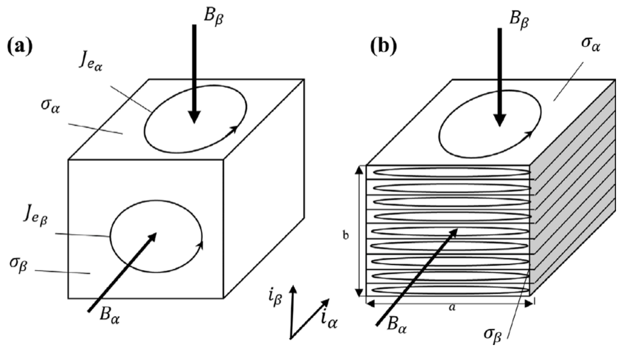

Fig. 4. Cross-section of a laminated nanocrystalline magnetic shielding layer. (a) Solid core model. (b) Laminated nanocrystalline model.

currents. Fig. 4 illustrates the cross-sections of different material magnetic shielding layers, where Fig. 4(a) represents a conventional solid model, and Fig. 4(b) represents a laminated nanocrystalline model. α and β denote the directions parallel and perpendicular to the shielding layer. As shown in Fig. 4(a), the electrical conductivity of conventional solid materials is typically measured using the four-point probe method [24], where the tested sample requires uniform thickness and surface connectivity of eddy currents ($J_{(e_\alpha)}$ and $J_{(e_\beta)}$). However, this method is not applicable to laminated nanocrystalline magnetic shielding layers because the interlayer insulation process interrupts the surface current flow between the nanocrystalline layers. As illustrated in Fig. 4(b), laminated nanocrystalline materials only interrupt the eddy currents in the thickness direction while allowing them to flow horizontally. Therefore, effective electrical conductivity differs in different directions.

In electric motors, laminated iron cores are commonly used to reduce eddy current losses. These operate on the same principle as laminated nanocrystalline materials, reducing magnetic noise by suppressing eddy currents. The issue of eddy currents in laminated iron cores is typically addressed by introducing anisotropic effective electrical conductivity for analytical calculations. Anisotropic electrical conductivity is currently utilized for analyzing eddy current losses in laminated magnetic cores [25] and in applications like inductive power transfer (IPT) [26]. It has been shown to provide more accurate results than isotropic models [27,28]. Additionally, the calculation of Johnson noise in nanocrystalline magnetic shielding devices aligns with the analysis of eddy current losses in laminated magnetic cores, further validating this method. The formula for the electrical conductivity parallel to the laminations is given by Eq. (7) [29]. In contrast, the formula for the effective anisotropic electrical conductivity in the perpendicular direction of the laminations is given by Eq. (8) [30].

$$\sigma_\alpha = e \cdot \sigma \tag{7}$$

$$\sigma_\beta = \frac{1}{e}\left(\frac{t}{a}\right)^2 \sigma \tag{8}$$

where σ represents the conductivity of a single nanocrystalline material, σ_α and σ_β represent the conductivities in the parallel and perpendicular directions, respectively, concerning the laminations. b denotes the cross-sectional width of the laminated nanocrystalline material, and t represents the thickness of a single nanocrystalline layer.

The magnetic field can be analyzed in three directions (B_x, B_y, B_z). Due to the geometric symmetry of the square magnetic shielding, the eddy currents generated by the magnetic fields in these three directions exhibit equivalence. As shown in Fig. 5, when the magnetic field in the X-direction (B_x) is projected onto the square magnetic shielding device, two faces have magnetic fields in the perpendicular direction (indicated in red), and four faces have magnetic fields in the parallel direction (indicated in blue), corresponding to the effective conductivities σ_α and σ_β, respectively. Due to the limited symmetry of the rectangular structure, simplifications need to be made based on the geometric structure. In this case, we will only focus on the analysis of the effective conductivity in the case of a cube. In subsequent sections, this effective conductivity model will be applied to establish the current Johnson noise model related to the nanocrystalline magnetic shielding box and eddy current losses. The model will be compared with actual magnetic noise testing and analysis.

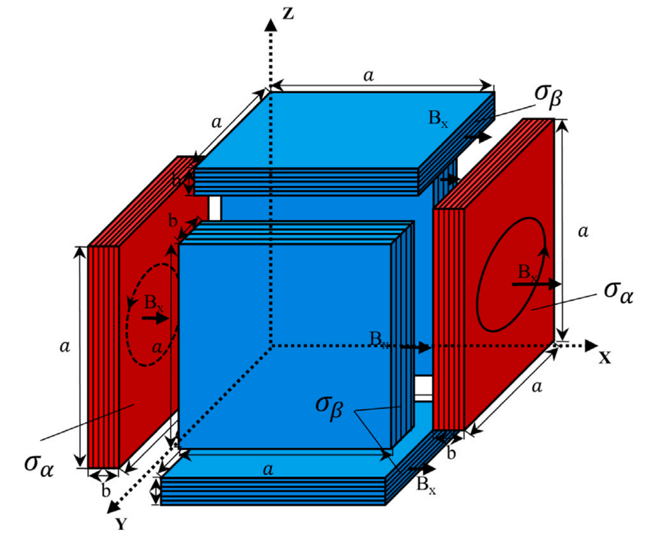

Fig. 5. The distribution of magnetic field components in the x direction on different faces of the laminated nanocrystalline cube.

3. The shielding factor considering lamination coefficient

In comparison to their larger counterparts, miniaturized biological magnetic shielding chambers offer the advantages of cost-effectiveness, shorter construction periods, and ease of portability. The cubic structure of the magnetic shielding box exhibits improved overall connectivity, exceptional shielding performance, and a more regular attenuation of residual magnetism and magnetic noise. Therefore, the assessment target of the magnetic shielding system in this paper is the cubic structure magnetic shielding box.

3.1. The model of shielding factor

The key determinant of the shielding effectiveness in stacked nanocrystalline cubes lies in the effective magnetic permeability. We comprehensively assessed nanocrystalline annulus with varying stacking coefficients and obtained an accurate model depicting the complex magnetic permeability as a function of stacking coefficient variations. Consequently, based on the nanocrystalline effective magnetic permeability model, we can establish a computational model for the nanocrystalline shielding coefficient under different stacking coefficients. The shielding coefficient is defined as the ratio between the external magnetic field strength H_0 under no shielding conditions and the internal magnetic field strength H at the same point under shielding conditions [31]. Eq. (9) represents the formula for the shielding coefficient of a single-layer spherical shield:

$$S_{\text{Sphere}} = \frac{H_0}{H} = 1 + \frac{2}{9} \cdot \frac{(\mu - 1)^2}{\mu} \cdot \left(1 - \frac{R_2^3}{R_1^3}\right) \tag{9}$$

where S_{Sphere} represents the relative magnetic permeability, μ represents the outer diameter, R_1 and R_2 represents the inner diameter of the sphere. Simplification of Eq. (9) yields:

$$S_{\text{sphere}} = 1 + \frac{2}{3} \cdot \mu \cdot \frac{b}{R_1} \tag{10}$$

Table 2
Magnetic field noise from high-permeability and nonmagnetic plate and shells of conductivity.

Simulation parameter condition	
Total thickness	2 mm
Length of side	250 mm
Electrical conductivity	$(1/1.2)\times 10^6$
Relative magnetic permeability of a single nanocrystalline layer	37 000
Number of nanocrystalline layers	100
Number of adhesive layers	99
Coil current	38.3 A

For square shielding, when the effective radius is taken to be half the length of the side, the equation for a cube can be derived from the spherical Eq. (11):

$$S_{\mathrm{Square}} \cong 1 + 2\cdot\lambda\cdot\mu\cdot\frac{b}{L_1} \tag{11}$$

where L_1 represents the outer side length of the cube containing the magnetic shielding layer. Assuming that the coefficient λ for the cube formula is similar to the coefficient for the cylindrical formula, we can set the coefficient for the cube λ to 0.48 within the acceptable range of error [32]. Therefore, the empirical formula for the shielding coefficient of a single-layer cube can be defined as:

$$S_{\mathrm{Square}} \cong 1 + 0.96\cdot\mu\cdot\frac{b}{L_1} \tag{12}$$

For the nanocrystalline layered structure, the magnetic permeability of the magnetic shielding layer varies with the stacking coefficient due to interlayer insulation effects. We have obtained the effective magnetic permeability considering the stacking coefficient under weak magnetic field conditions. Therefore, to more accurately simulate the impact of the stacking coefficient on the shielding performance, we can substitute the fitted magnetic permeability Eq. (5) into Eq. (12) to obtain the shielding coefficient of a single-layer nanocrystalline cube:

$$S_{\mathrm{Nanocry}} \cong 1 + (35846.4\cdot e - 814.8)\cdot\frac{b}{L_1} \tag{13}$$

3.2. The simulation of shielding factor

The residual magnetization at the center is a criterion for evaluating the shielding effectiveness of a magnetic shielding device. By using Finite Element Method (FEM) calculations to determine the residual magnetization inside the magnetic shielding device, we can numerically simulate the shielding coefficient of the cube. Traditional modeling methods have primarily focused on conventional dense solid models, necessitating improvements for multi-layer stacked nanocrystalline magnetic shielding devices. In this article, we address the unique interlayer insulation and stacked structure of nanocrystalline materials by introducing multi-layer transition boundary conditions in the finite element simulation. This modeling method is suitable when the thickness of the conductor is comparable to the skin depth.

The arrangement of the multi-layer stacked magnetic shielding layers follows the physical layout of the nanocrystalline magnetic shielding box. The material is set as a stack of multiple layers, each consisting of alternating nanocrystalline and adhesive. The boundary condition module is designed to incorporate the multi-layer transition boundary conditions. As the transition boundary conditions can only be calculated in the frequency domain, a frequency of 0.1 Hz is chosen to calculate static magnetic shielding. As presented in Table 2, the total thickness of the stacked nanocrystalline cube is set to 2 mm, with a side length of 250 mm. The electrical conductivity is $(1/1.2)\times 10^6$ and the relative permittivity is 1. In actual testing, the relative magnetic permeability of a single nanocrystalline layer without interlayer insulation conditions is found to be 37 000. The number of nanocrystalline layers is set to 100, while the number of adhesive layers is set to 99. The stacking coefficient, denoted as e, represents the ratio of the actual thickness of the nanocrystalline layer to the total thickness of the stacked nanocrystalline cube, while keeping the total thickness of the magnetic shielding layer unchanged at 2 mm. By varying the actual thickness of the nanocrystalline layer, the size of the stacking coefficient can be adjusted accordingly. For the simulation of the shielding coefficient, a uniform magnetic field is generated by a pair of parallel and connected coaxial square coils (Helmholtz coils). These coils have a length of 1350 mm and a width of 750 mm and are separated by a distance of 500 mm. The current is set to 38.3 A, resulting in a magnetic flux density of approximately 50 000 nT at the center point.

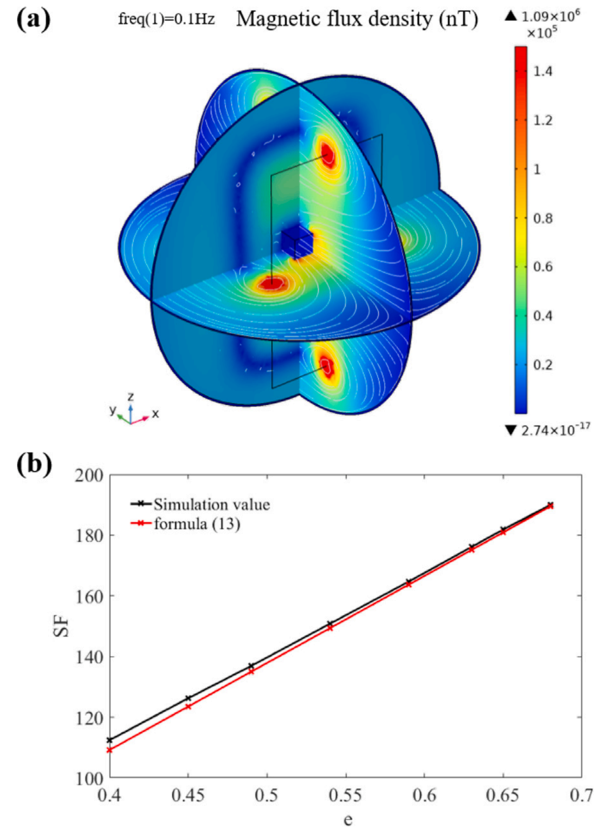

Fig. 6. Simulation of nanocrystalline magnetic shielding box with different stacking coefficients. (a) Magnetic flux density distribution of nanocrystalline cube under uniform magnetic field. (b) Comparison of simulated and calculated shielding coefficients of magnetically shielded boxes with different stacking coefficients.

The internal magnetic field distribution within the nanocrystalline cube is depicted in Fig. 6(a). The shielding coefficient of the FEM simulation can be determined by evaluating the residual magnetization at the center of the stacked nanocrystalline magnetic shielding box for different stacking coefficients. As depicted in Fig. 6(b), both the calculation from Eq. (13) and the finite element method (FEM) simulation yield linear curves for the shielding factor (SF), with the difference between the two results not exceeding 5%. These two approaches mutually corroborate the linearizing effect of the stacking coefficient on the shielding factor. Therefore, Eq. (13) can be used to describe the trend of the shielding coefficient variation in the stacked nanocrystalline cube magnetic shielding structure to a certain extent.

4. The magnetic noise considering the stacking effect

4.1. The analysis of magnetic noise

The utilization of laminated nanocrystalline magnetic shielding devices allows for the implementation of interlayer insulation techniques to suppress eddy currents and improve eddy current losses, thereby reducing internal noise. Therefore, it is necessary to refine the magnetic noise model for laminated nanocrystalline cubic structures.

According to the fluctuation–dissipation theory [33], magnetic noise can be calculated based on the energy dissipation in the material. By relating the magnetic noise to the power dissipation through the energy spectral density, the complex calculation of magnetic noise can be transformed into solving for the dissipation. According to the Nyquist relation, the magnetic field noise spectrum at a certain point inside a dissipative magnetic shield can be calculated based on the average power dissipation generated by the oscillating current flowing through the hypothetical excitation coil located at the same point within the shield material. Calculating the electric field on the inner surface of the conductive shell caused by the electric dipoles yields the Johnson current noise and thermal magnetization noise of an infinitely large plate, as shown in Eqs. (14) and (15) [34].

$$\delta B_{\text{curr}} = \frac{1}{\sqrt{6\pi}} \cdot \frac{\mu_0 \sqrt{kTd\sigma_m}}{r} \tag{14}$$

$$\delta B_{\text{magn}} = \frac{1}{2\pi r} \cdot \sqrt{\frac{\mu_0 kT}{fd}} \cdot \sqrt{\frac{\mu''}{\mu'^2}} \tag{15}$$

where μ_0 represents the vacuum permeability, k is the Boltzmann constant, T denotes the material temperature, σ_m represents the material conductivity, f represents the frequency, and d indicates the effective thickness of the material, and r represents the vertical distance to the shielding layer.

From Eq. (14), it can be inferred that the Johnson current noise is directly proportional to the square of the equivalent conductivity of the magnetic shielding layer. Due to the variation in the truncation of eddy currents in different directions caused by the laminating process, Fig. 5 illustrates the distribution of the equivalent conductivity of the nanocrystalline square magnetic shielding device, denoted as σ_α and σ_β. Due to the geometric symmetry of the square magnetic shield, the magnetic noise induced by the magnetic fields in three directions is equivalent. Hence, the equivalent conductivity of two of its faces can be set as σ_α, and the equivalent magnetic conductivity of the other four faces can be set as σ_β. By substituting Eqs. (7) and (8) into Eq. (14), the magnitude of Johnson noise in the laminated nanocrystalline square magnetic shielding device can be derived as expressed in Eq. (16). Likewise, according to Eq. (15), the thermal magnetization noise is directly proportional to the square of the imaginary magnetic conductivity of the shielding layer and inversely proportional to its real magnetic conductivity. Taking into account the unique pattern of variation in the complex magnetic conductivity of the nanocrystalline material with the lamination coefficient, Eqs. (5) and (6) can be substituted into Eq. (15), resulting in the expression for the magnitude of thermal magnetization noise in the laminated nanocrystalline square magnetic shielding device as given by Eq. (17).

$$\delta B_{\text{curr}} = \frac{2\mu_0}{r} \cdot \sqrt{\frac{kTe\sigma d}{6\pi}} + \frac{4\mu_0 t}{ar} \cdot \sqrt{\frac{kTd\sigma}{6\pi e}} \tag{16}$$

$$\delta B_{\text{magn}} = \frac{3}{\pi r} \cdot \sqrt{\frac{\mu_0 kT}{fd}} \cdot \sqrt{\frac{214.6 \cdot e - 16}{(37340 \cdot e - 848.8)^2}} \tag{17}$$

$$\delta B = \delta B_{\text{curr}} + \delta B_{\text{magn}} \tag{18}$$

Based on Eq. (18), the calculated values of the total noise in the iron-based nanocrystalline cubic structure can be obtained by varying the lamination coefficient. As illustrated in Fig. 7, at lower frequencies (0.1–10 Hz), the low-frequency total noise gradually decreases as the lamination coefficient increases. As the frequency increases, although the total noise tends to increase with the lamination coefficient, the trend is mild and not pronounced. As depicted in Fig. 6(b), it is evident that the shielding coefficient increases with the lamination coefficient. Therefore, selecting materials with higher lamination coefficients for practical applications is advisable to enhance shielding effectiveness while reducing magnetic noise.

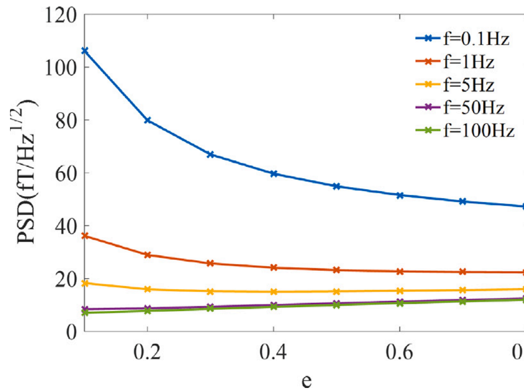

Fig. 7. The trend of total noise with varying stacking factors at different frequencies.

4.2. The magnetic noise test system

A cubic magnetic shielding box consisting of laminated nano-crystalline entities has been constructed for practical measurements, with a side length of 250 mm, a 12 mm thick magnetic shielding layer, and a stacking factor of e = 0.6. The outer structure is made of acrylic transparent panels. Iron-based nano-crystalline materials are uniformly affixed to the interior surface of the cubic acrylic transparent panels, with a single layer thickness of 20 μm and a double-sided adhesive thickness of 10 μm. After affixing, the entire shell is cured and shaped. Through-cut optical holes are arranged on the front and back of the box, created by circular blades and sealed with non-magnetic polyethylene insulating material at the blade positions to isolate external wear.

The aim of this study is to examine the magnetic noise of nanocrystalline materials. However, during the testing process, the magnetic noise of the materials is easily masked by interferences from the external environment. To more accurately assess the intrinsic noise level of the materials, we conducted magnetic noise tests by placing the nanocrystalline magnetic shielding box inside a magnetic shielding room and a double-layer Permalloy scaling model. This effectively shielded most of the external environmental noise, thereby achieving precise measurement of the material's magnetic noise. Consequently, the testing results of the magnetic field measurement device (SERF magnetometer) primarily reflect the magnetic noise generated by the innermost layer of the nanocrystalline magnetic shielding box, when the high-shielding coefficient Permalloy magnetic shielding device effectively suppresses the external environmental magnetic noise interference. The shielding coefficients of the double-layer Permalloy with nanocrystalline magnetic shielding box at different frequencies are shown in Table 3. The measurement results indicate that the shielding coefficients are generally constant in the frequency range of 0.1–95 Hz. The shielding coefficient slightly decreases at low frequencies and increases with increasing frequency. It is important to note that due to power frequency interference, we did not measure the shielding coefficients at 50 Hz and 100 Hz frequencies.

As depicted in Fig. 8(a), the magnetic field measurement device (SERF magnetometer) was positioned at the center of the iron-based nanocrystalline magnetic shielding box. The power spectral density of the sensor's magnetic noise is shown in Fig. 8(b). The testing was conducted in a temperature-stabilized underground chamber at 280.15 K during nighttime, ensuring a stable environmental noise level. When the high shielding coefficient Permalloy magnetic shielding device effectively suppressed the external environmental magnetic noise interference, the test results of the magnetic field measurement device primarily revealed the magnetic noise generated by the innermost layer of the nanocrystalline magnetic shielding box. From Fig. 8(b), it can be observed that in the frequency range of 10 to 100 Hz the

Table 3
Shielding coefficients of 2-layer Permalloy with nanocrystalline magnetic shielding box at different frequencies.

Frequency	X-position	Y-position	Z-position
0.1	465 ± 1.3	258 ± 2.4	432 ± 1.3
1	513 ± 2.4	287 ± 1.6	486 ± 1.5
5	898 ± 1.75	768 ± 1.4	912 ± 1.3
10	1470 ± 0.3	1390 ± 1.35	1367 ± 1.2
20	2409 ± 1.4	2334 ± 0.5	2506 ± 2.3
30	3212 ± 1.9	3112 ± 1.3	3123 ± 1.3
40	3854 ± 1.3	3439 ± 1.73	3780 ± 1.8
60	5256 ± 2.3	5027 ± 1.8	5363 ± 1.3
70	6195 ± 1.4	5941 ± 1.6	5906 ± 2.8
80	6671 ± 2.6	6261 ± 2.3	6543 ± 1.6
90	7227 ± 1.5	6535 ± 1.3	7084 ± 1.3
95	7541 ± 1.6	7378 ± 1.9	7346 ± 1.7

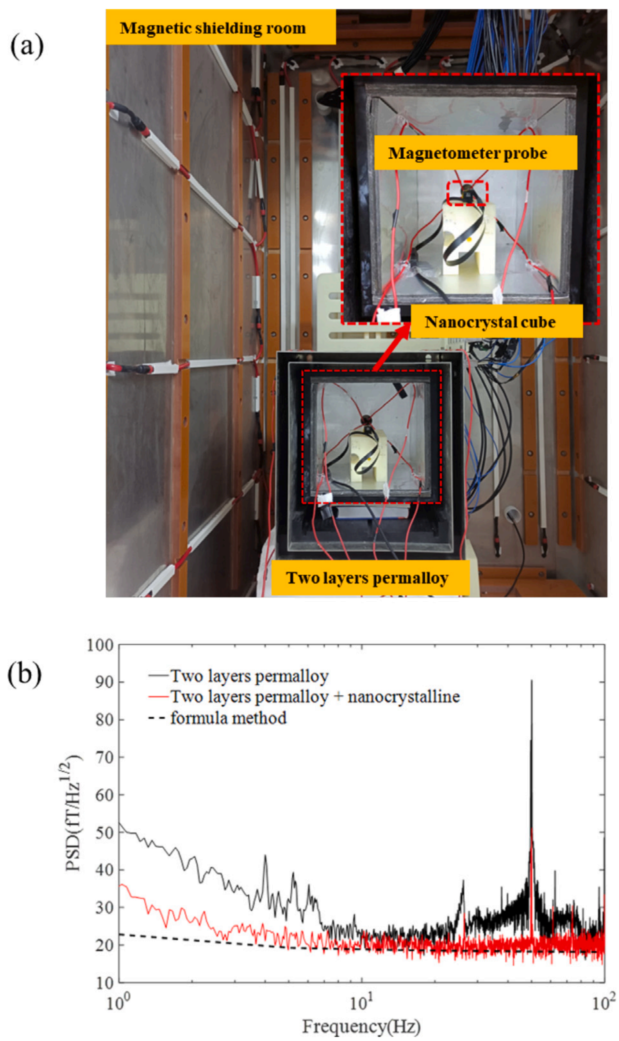

Fig. 8. Magnetic noise as a function of frequency test for iron-based nanocrystalline magnetic shielding boxes with different thicknesses. (a) Magnetic noise testing environment for iron-based nanocrystalline. (b) Comparison between the measured magnetic noise of the iron-based nanocrystalline magnetic shielding box and the total noise calculated by the equation.

magnetic noise of the nanocrystalline magnetic shielding box remained stable, which can be defined as Johnson noise independent of frequency variation, while at frequencies below 10 Hz, the magnetic noise showed a clear trend of changing with frequency, which can be defined as thermal-magnetic noise.

By utilizing Eq. (18), we can calculate the thermal-magnetic and Johnson noise of a 12 mm thick laminated nanocrystalline cubic sample with a stacking factor of 0.6 and a side length of 250 mm. The comparison between the measured values and the total noise calculated using the formula is shown in Fig. 8(b). In the frequency range of 10 to 100 Hz, the Johnson current noise, which is uncorrelated with frequency variations, remains constant. However, at frequencies below 10 Hz, the low shielding effectiveness of the iron-based nanocrystalline magnetic shield box results in incomplete shielding of some Permalloy magnetic noise and environmental noise. This interference error causes the measured values in the low-frequency range to be slightly higher than the calculated magnetic noise from the formula.

5. Conclusion

In summary, this paper presents a methodology for the quantitative evaluation and investigation of the shielding factor and magnetic noise in a nanocrystalline magnetic shielding system, considering the effects of lamination. By examining the complex permeability of nanocrystalline test rings with different lamination factors, we summarize a model of equivalent permeability that varies with the lamination factor and incorporate it into the shielding factor formula. Furthermore, through the introduction of a multilayer transition boundary condition module, we have refined the finite element method (FEM) for nanocrystalline magnetic shielding devices, facilitating the simulation of spatial magnetic field distribution under the influence of lamination, and compared it with the proposed shielding factor model, verifying their effectiveness with a margin of error within 5%. Lastly, considering the model of equivalent conductivity in different directions due to lamination, we have analyzed and enhanced the magnetic noise calculation model applicable to square nanocrystalline magnetic shields and validated the precision of the model through experimentation.

CRediT authorship contribution statement

Ling Wang: Conceptualization, Methodology, Resources, Software, Writing – original draft, Writing – review & editing, Visualization. **Minxia Shi:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Funding acquisition. **Yun Le:** Funding acquisition, Investigation, Resources, Supervision, Writing – review & editing. **Xu Zhang:** Investigation, Supervision, Writing – review & editing. **Jianzhi Yang:** Methodology, Writing – original draft, Writing – review & editing. **Shuai Yuan:** Methodology, Visualization, Writing – review & editing. **Yuzheng Ma:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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